상태 관리

권기웅

1. Context API

1.1 context 활용 준비

1.1.1 context

kwonDivContext.js

```
const { createContext } = require("react");

const KwonDivContext = createContext({
  bg: "red",
  txt: "R",
});

export default KwonDivContext; // exp
```

1.1.2 consumer

kwonDiv.js

```
import React, { useContext } from "react";
import KwonDivContext from "./kwonDivContext";

const KwonDiv = () => {
  const kdc = useContext(KwonDivContext);
  return (
    <div
      style={{
        width: "100px",
        height: "100px",
        backgroundColor: kdc.bg,
        color: "white",
        fontSize: "20pt",
      }}
    >
      {kdc.txt}
    </div>
  );
};

export default KwonDiv;
```

1.1.3 App.js

App.js

```
import "./App.css";
import KwonDiv from "./kwonDiv";

function App() {
  return <KwonDiv />;
}

export default App;
```

1.2 provider

1.2.1 context

kwonDivContext.js

```
const { createContext } = require("react");

const KwonDivContext = createContext();

export default KwonDivContext;
```

1.2.2 App.js

App.js

```
import "./App.css";
import KwonDiv from "./kwonDiv";
import KwonDivContext from "./kwonDivContext";

function App() {
  return (
    <KwonDivContext.Provider value={{ bg: "red", txt: "빨" }}>
      <KwonDiv />
    </KwonDivContext.Provider>
  );
}

export default App;
```

1.3 context 수정하게

kwonDivContext.js

```
const { createContext, useState } = require("react");

const KwonDivContext = createContext();

export const KwonDivProvider = ({ children }) => {
  const [v, setV] = useState({ bg: "red", txt: "hi" });

  return (
    <KwonDivContext.Provider
      value={{
        bg: v.bg,
        txt: v.txt,
        changeee: setV,
      }}
    >
      {children}
    </KwonDivContext.Provider>
  );
};

export default KwonDivContext;
```

kwonBtns.js

```
import React, { useContext } from "react";
import KwonDivContext from "./kwonDivContext";

const KwonBtns = () => {
  const kdc = useContext(KwonDivContext);
  return (
    <>
      <button onClick={() => { kdc.changeee({ bg: "red", txt: "빨" }); }}>
        R
      </button>
      <button onClick={() => { kdc.changeee({ bg: "green", txt: "녹" }); }}>
        G
      </button>
      <button onClick={() => { kdc.changeee({ bg: "blue", txt: "파" }); }}>
        B
      </button>
    </>
  );
};

export default KwonBtns;
```

App.js

```
import "./App.css";
import KwonBtns from "./kwonBtns";
import KwonDiv from "./kwonDiv";
import { KwonDivProvider } from "./kwonDivContext";

function App() {
  return (
    <KwonDivProvider>
      <KwonDiv />
      <KwonBtns />
    </KwonDivProvider>
  );
}

export default App;
```

2. redux

2.1 라이브러리 추가

yarn add @reduxjs/toolkit react-redux

```
Windows PowerShell
D:₩Kwon₩wsReact₩r12_state2>yarn add @reduxjs/toolkit react-redux
yarn add v1.22.19
[1/4] Resolving packages...
[2/4] Fetching packages...
[3/4] Linking dependencies...
warning " > @testing-library/user-event@13.5.0" has unmet peer dependency "@testing-library/dom@>=7.21.4".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer dependency "@babel/plugin-syntax-
flow@^7.14.5".
warning "react-scripts > eslint-config-react-app > eslint-plugin-flowtype@8.0.3" has unmet peer dependency "@babel/plugin-transfo
rm-react-jsx@^7.14.9".
warning "react-scripts > react-dev-utils > fork-ts-checker-webpack-plugin@6.5.2" has unmet peer dependency "typescript@>= 2.7".
warning "react-scripts > eslint-config-react-app > @typescript-eslint/eslint-plugin > tsutils@3.21.0" has unmet peer dependency "
typescript@>=2.8.0 || >= 3.2.0-dev || >= 3.3.0-dev || >= 3.4.0-dev || >= 3.5.0-dev || >= 3.6.0-dev || >= 3.6.0-beta || >= 3.7.0-d
ev || >= 3.7.0-beta".
[4/4] Building fresh packages...
success Saved lockfile.
success Saved 9 new dependencies.
info Direct dependencies
├─@reduxjs/toolkit@1.8.6
└─react-redux@8.0.4
info All dependencies
├─@reduxjs/toolkit@1.8.6
├─@types/hoist-non-react-statics@3.3.1
├─@types/use-sync-external-store@0.0.3
├─hoist-non-react-statics@3.3.2
├─react-redux@8.0.4
├─redux-thunk@2.4.1
├─redux@4.2.0
├─reselect@4.1.6
└─use-sync-external-store@1.2.0
Done in 5.33s.
```

2.2 일단은

kwonH1.js

```python
import React from "react";

const KwonH1 = () => {
  return <h1>일단은</h1>;
};

export default KwonH1;
```

App.js

```python
import "./App.css";
import KwonH1 from "./kwonH1";

function App() {
  return <KwonH1 />;
}

export default App;
```

2.3 폰트사이즈 변경

2.3.1 reducer

resizerSlice.js

```
// rxslice
import { createSlice } from "@reduxjs/toolkit";

const initialState = { size: 20 };

export const resizerSlice = createSlice({
  name: "rs",
  initialState,
  reducers: {
    bigger: (state) => {
      state.size = state.size <= 100 ? state.size + 10 : state.size;
    },
    smaller: (st) => {
      st.size = st.size >= 20 ? st.size - 10 : st.size;
    },
  },
});

export const { bigger, smaller } = resizerSlice.actions;

export default resizerSlice.reducer;
```

2.3.2 store

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import "./index.css";
import App from "./App";
import reportWebVitals from "./reportWebVitals";
import { configureStore } from "@reduxjs/toolkit";
import resizerSlice from "./resizerSlice";
import { Provider } from "react-redux";

const store = configureStore({
  reducer: {
    rs: resizerSlice,
  },
});

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <Provider store={store}>
    <App />
  </Provider>
);

// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();
```

2.3.3 subscriber

kwonH1.js

```
import React from "react";
import { useSelector } from "react-redux";

const KwonH1 = () => {
  const sizeee = useSelector((state) => state.rs.size);
  return <h1 style={{ fontSize: sizeee + "pt" }}>일단은</h1>;
};

export default KwonH1;
```

2.3.4 dispatcher

kwonResizeBtns.js

```
import React from "react";
import { useDispatch } from "react-redux";
import { bigger, smaller } from "./resizerSlice";

const KwonResizeBtns = () => {
  const d = useDispatch();

  return (
    <>
      <button
        onClick={() => {
          d(smaller());
        }}
      >
        작
      </button>
      <button
        onClick={() => {
          d(bigger());
        }}
      >
        커
      </button>
    </>
  );
};

export default KwonResizeBtns;
```

2.3.5 App.js

App.js

```
import "./App.css";
import KwonH1 from "./kwonH1";
import KwonResizeBtns from "./kwonResizeBtns";

function App() {
  return (
    <>
      <KwonResizeBtns />
      <hr />
      <KwonH1 />
    </>
  );
}

export default App;
```

2.4 하나 더(내용 변경)

2.4.1 reducer

txtSlice.js

```
import { createSlice } from "@reduxjs/toolkit";

const initialState = {
  txt: "시간당 100만원",
};

export const txtSlice = createSlice({
  name: "ts",
  initialState,
  reducers: {
    changee: (s, actionnn) => {
      s.txt = actionnn.payload;
    },
  },
});

export const { changee } = txtSlice.actions;

export default txtSlice.reducer;
```

2.4.2 store

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import "./index.css";
import App from "./App";
import reportWebVitals from "./reportWebVitals";
import { configureStore } from "@reduxjs/toolkit";
import resizerSlice from "./resizerSlice";
import { Provider } from "react-redux";
import txtSlice from "./txtSlice";

const store = configureStore({
  reducer: {
    rs: resizerSlice,
    ts: txtSlice,
  },
});

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <Provider store={store}>
    <App />
  </Provider>
);

// If you want to start measuring performance in your app, pass a function
// to log results (for example: reportWebVitals(console.log))
// or send to an analytics endpoint. Learn more: https://bit.ly/CRA-vitals
reportWebVitals();
```

2.4.3 subscriber

kwonH1.js

```
import React from "react";
import { useSelector } from "react-redux";

const KwonH1 = () => {
  const sizeee = useSelector((state) => state.rs.size);
  const t = useSelector((s) => s.ts.txt);
  return <h1 style={{ fontSize: sizeee + "pt" }}>{t}</h1>;
};

export default KwonH1;
```

2.4.4 dispatcher/subscriber

kwonTxtInput.js

```
import React from "react";
import { useDispatch, useSelector } from "react-redux";
import { changee } from "./txtSlice";

const KwonTxtInput = () => {
  const v = useSelector((s) => s.ts.txt);
  const d = useDispatch();

  return (
    <input
      value={v}
      onChange={(e) => {
        d(changee(e.target.value));
      }}
    />
  );
};

export default KwonTxtInput;
```

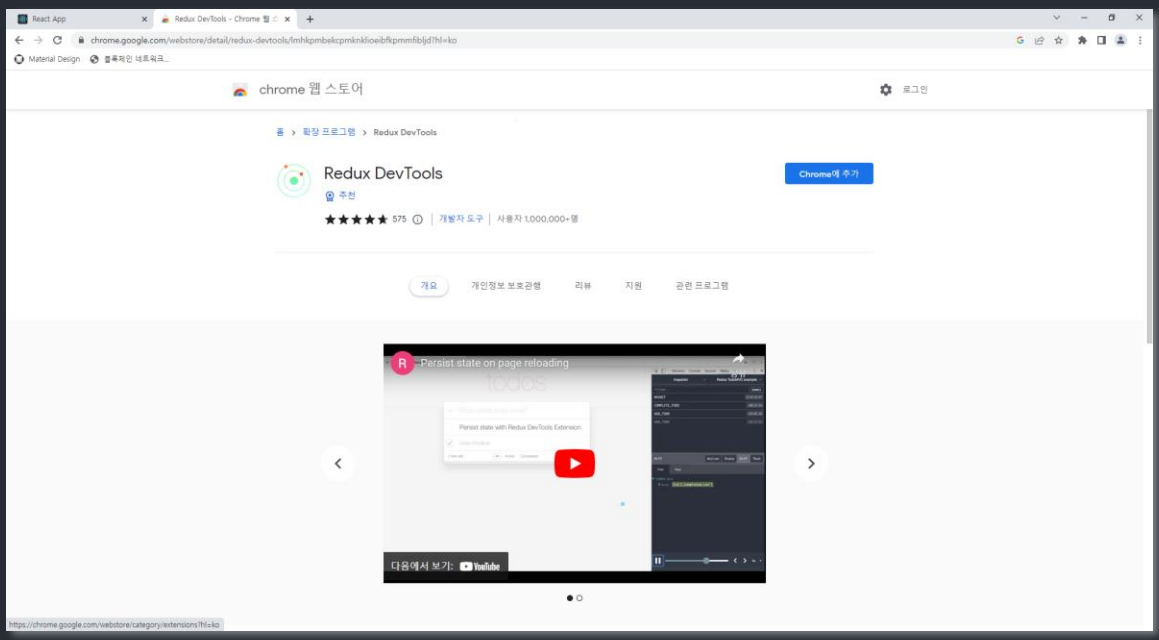
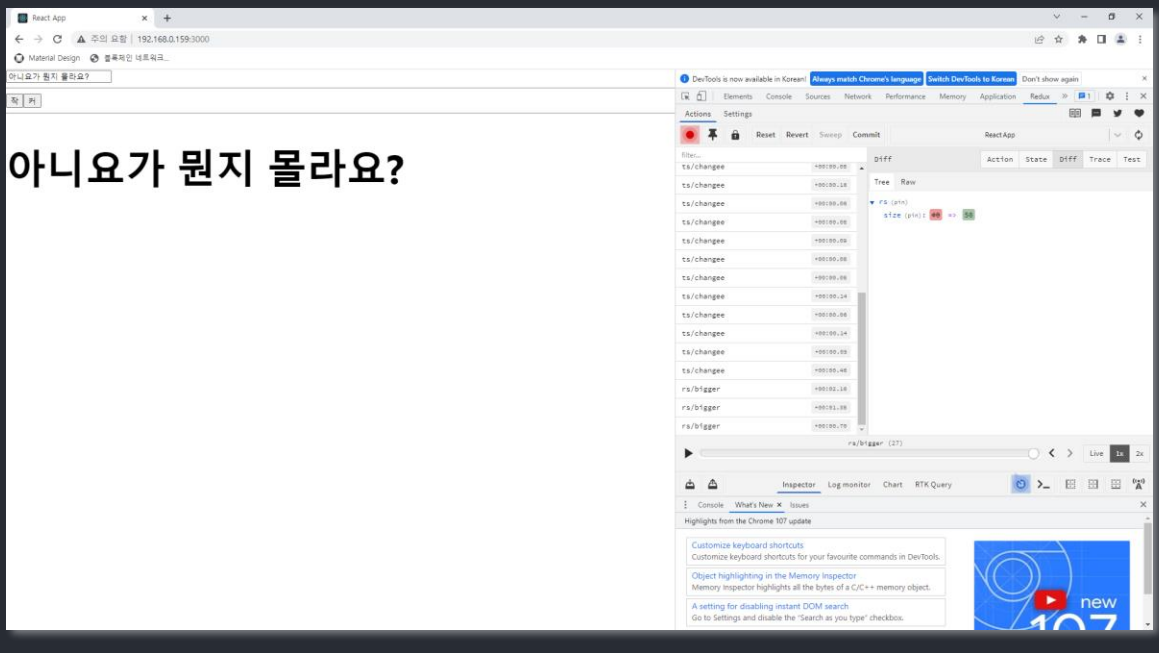
2.4.5 App.js

App.js

```
import "./App.css";
import KwonH1 from "./kwonH1";
import KwonResizeBtns from "./kwonResizeBtns";
import KwonTxtInput from "./kwonTxtInput";

function App() {
  return (
    <>
      <KwonTxtInput />
      <hr />
      <KwonResizeBtns />
      <hr />
      <KwonH1 />
    </>
  );
}

export default App;
```

2.5 chrome 확장프로그램

Redux DevTools